

TECHNICAL INTERVIEW SERIES

DevOps Interview Prep Guide

12 essential questions and answers to help you
walk into your next DevOps interview with confidence.

12

Questions

06

Core Topics

100%

Interview Ready

TOPICS COVERED

- Introduction to DevOps
- Containers & Orchestration
- Monitoring & Deployment Strategies
- Version Control & Infrastructure as Code
- CI/CD Pipelines
- Scaling & Networking

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Introduction to DevOps

Q1

What is DevOps, and why is it important?

DevOps is a set of practices that combines software development (Dev) and IT operations (Ops) to shorten the software development lifecycle while delivering high-quality software continuously. It matters because it promotes collaboration, automation, and rapid iteration between teams that once worked in silos.

Q2

What is the difference between Continuous Integration and Continuous Deployment?

Continuous Integration (CI) is the practice of merging all developers' code into a shared repository frequently, followed by automated builds and tests. Continuous Deployment (CD) extends this by automatically releasing every change that passes testing straight to production.

Practice tip: say the answer out loud before you check it.



Version Control & Infrastructure as Code

Q1

What is the purpose of a Version Control System (VCS)?

A VCS tracks changes to code over time, enables collaboration among multiple developers, maintains a full history of revisions, and allows reverting to earlier versions when needed. Git is the most widely used VCS today.

Q2

Explain the concept of Infrastructure as Code (IaC).

IaC is the practice of managing and provisioning infrastructure through machine-readable definition files instead of manual processes. It enables consistent, repeatable deployments and version-controlled infrastructure changes, using tools such as Terraform and CloudFormation.

Practice tip: say the answer out loud before you check it.



Containers & Orchestration

Q1

What is a container, and how does it differ from a virtual machine?

A container packages an application with its dependencies while sharing the host OS kernel, making it lightweight and fast to start. A virtual machine includes a full guest OS, making it heavier and slower to boot but offering stronger isolation.

Q2

What is Docker, and why is it popular in DevOps?

Docker is a platform for developing, shipping, and running applications inside containers. It's popular because it guarantees consistency across environments, simplifies dependency management, and integrates smoothly with CI/CD pipelines.

Practice tip: say the answer out loud before you check it.



Q1

What is Kubernetes, and what problem does it solve?

Kubernetes is a container orchestration platform that automates the deployment, scaling, and management of containerized applications. It solves the challenge of running many containers across multiple hosts, handling failover, load balancing, and automatic scaling.

Q2

What is a CI/CD pipeline?

A CI/CD pipeline is an automated sequence of steps that takes code from commit to production - building, testing, and deploying the application - reducing manual effort and the risk of human error.

Practice tip: say the answer out loud before you check it.



Monitoring & Deployment Strategies

Q1

What is the difference between monitoring and logging?

Monitoring tracks system metrics such as CPU, memory, and response time in real time to detect issues as they happen. Logging captures detailed event records used for debugging and auditing. Both are essential pillars of observability.

Q2

What is a blue-green deployment strategy?

Blue-green deployment maintains two identical production environments. Live traffic is routed to one (blue) while the other (green) is updated and tested. Once verified, traffic switches over to green, minimizing downtime and risk.

Practice tip: say the answer out loud before you check it.



Scaling & Networking

Q1

What is the role of a reverse proxy in DevOps?

A reverse proxy sits in front of application servers, forwarding client requests to the appropriate backend while handling load balancing, SSL termination, and caching. NGINX and HAProxy are common examples.

Q2

What is the difference between horizontal and vertical scaling?

Horizontal scaling adds more machines or instances to handle load, improving fault tolerance. Vertical scaling adds more resources - CPU, RAM - to an existing machine, which is simpler to do but has physical limits.

Practice tip: say the answer out loud before you check it.

